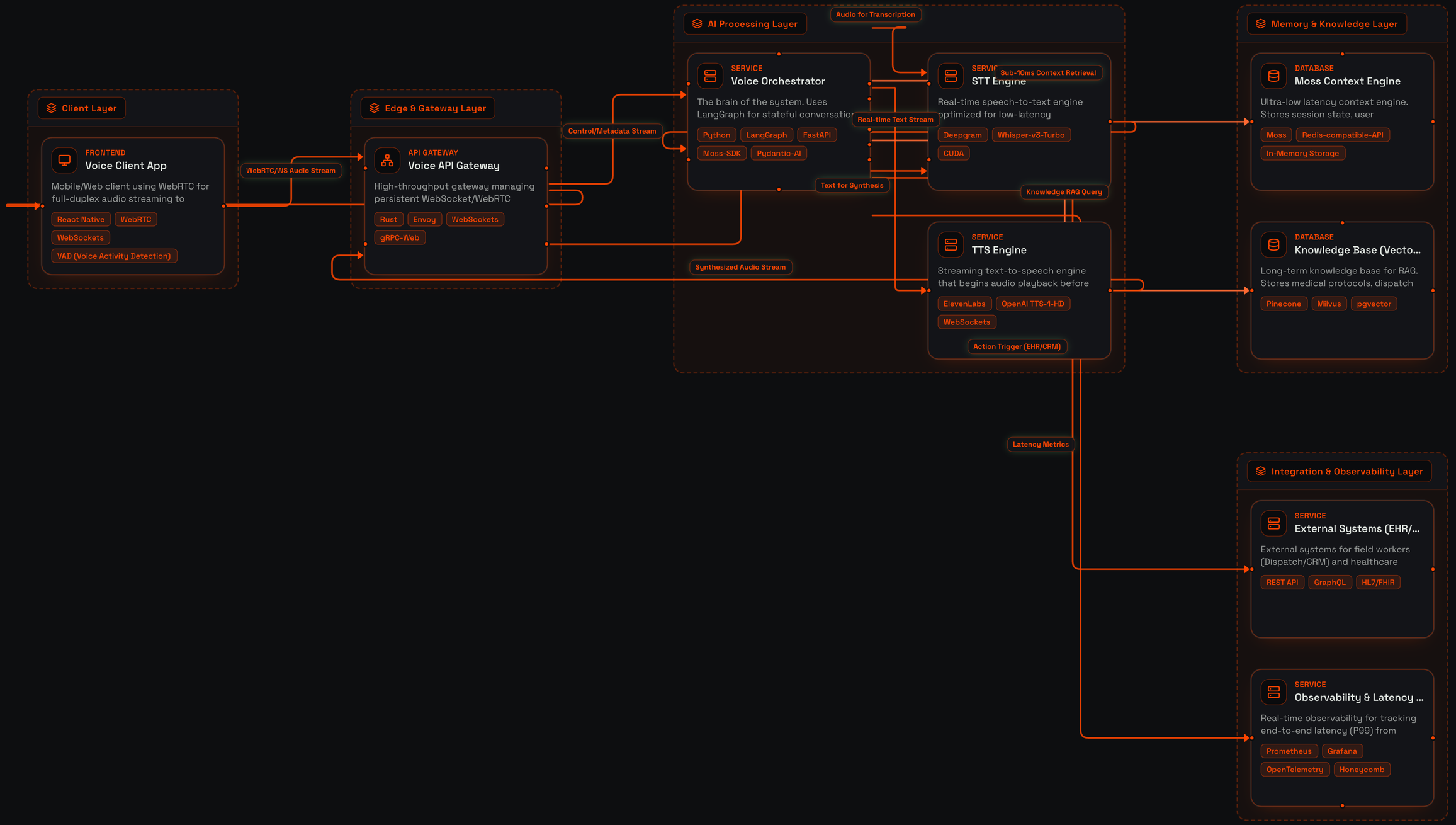
Client Layer
FRONTEND
Voice Client App
Mobile/Web client using WebRTC for full-duplex audio streaming to
React Native
WebRTC
WebSockets
VAD (Voice Activity Detection)
WebRTC/WS Audio Stream
Edge & Gateway Layer
API GATEWAY
Voice API Gateway
High-throughput gateway managing persistent WebSocket/WebRTC
Rust
Envoy
WebSockets
gRPC-Web
Control/Metadata Stream
AI Processing Layer
Audio for Transcription
SERVICE
Voice Orchestrator
The brain of the system. Uses LangGraph for stateful conversation
Python
LangGraph
FastAPI
Moss-SDK
Pydantic-AI
Real-time Text Stream
Text for Synthesis
SERVICE
STT Engine
Sub-10ms Context Retrieval
Real-time speech-to-text engine optimized for low-latency
Deepgram
Whisper-v3-Turbo
CUDA
Knowledge RAG Query
Synthesized Audio Stream
SERVICE
TTS Engine
Streaming text-to-speech engine that begins audio playback before
ElevenLabs
OpenAI TTS-1-HD
WebSockets
Action Trigger (EHR/CRM)
Memory & Knowledge Layer
DATABASE
Moss Context Engine
Ultra-low latency context engine. Stores session state, user
Moss
Redis-compatible-API
In-Memory Storage
DATABASE
Knowledge Base (Vecto...
Long-term knowledge base for RAG. Stores medical protocols, dispatch
Pinecone
Milvus
pgvector
Latency Metrics
Integration & Observability Layer
SERVICE
External Systems (EHR/...
External systems for field workers (Dispatch/CRM) and healthcare
REST API
GraphQL
HL7/FHIR
SERVICE
Observability & Latency ...
Real-time observability for tracking end-to-end latency (P99) from
Prometheus
Grafana
OpenTelemetry
Honeycomb